

Lab_Syn_2.0

You have to synthesize **ALU.v** file using **Design Compiler shell** and generate **Technology Dependent** gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside **std_cells** folder:-

- **scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db**
- **scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c.db**
- **scmetro_tsmc_cl013g_rvt_ss_1p08v_125c.db**

Steps: -

1. Copy **Lab_Syn_2.0** folder from your computer into the virtual machine inside Labs directory
 - Open a terminal inside path **/home/ICer/Labs/Lab_Syn_2.0/syn** and run **syn_script.tcl** script through design compiler shell using the following command in the terminal
 - **dc_shell -f syn_script.tcl > syn.log**
2. Every Time you synthesize a design, you need to check the synthesis log (**syn.log**) to be sure that: -
 - **No Errors**
 - **No Warnings**
 - **No Latches**
 - **No Comb Loops**

During Synthesize ALU.v using **syn_script.tcl**, you will find 4 issues, 2 of them will be solved by modifying in the **syn_script .tcl** and the other 2 issues needs modifications in **ALU.v**. Please fix the 4 issues and be sure that your synthesis log is free of Errors/Warnings, latches and loops.

Note: Can you find the rule of this command at the beginning of the script

define_design_lib work -path ./work